LOCALIZING A SPECIAL COCONTINUOUS FUNCTOR STAYS SPECIAL COCONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The five conditions characterizing an equivalence of topoi. Same chapter, Lemma lemma-equivalence. Supplies the five properties verified for the induced functor on the localized sites.
- The localization of a cocontinuous functor and its commutativity. Same chapter, Lemma lemma-localize-cocontinuous. Supplies the existence of the induced functor and the commutativity of the displayed diagrams.
- Continuity and cocontinuity. Same chapter, Definition definition-cocontinuous and the definition of a continuous functor. Supplies the behaviour of coverings and refinements under the functor.

Source reference. Original lemma 03EG: sources/stacks-sites/pinned-original-source.tex, source lines 7129–7129. Navigation pointer invoked at proof line 7129; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03A0: sources/stacks-sites/pinned-original-source.tex, source lines 7108–7108. Navigation pointer invoked at proof line 7108; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$, $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be a special cocontinuous functor. For every object U of $\mathcal{C}$ we have a commutative diagram*

$$\begin{array}{ccc} \mathcal{C}/U & \xrightarrow{j_U} & \mathcal{C} \\ \downarrow & & \downarrow u \\ \mathcal{D}/u(U) & \xrightarrow{j_{u(U)}} & \mathcal{D} \end{array}$$

as in Lemma 03EG. The left vertical arrow is a special cocontinuous functor. Hence in the commutative diagram of topoi

$$\begin{array}{ccc} Sh(\mathcal{C}/U) & \xrightarrow{j_U} & Sh(\mathcal{C}) \\ \downarrow & & \downarrow u \\ Sh(\mathcal{D}/u(U)) & \xrightarrow{j_{u(U)}} & Sh(\mathcal{D}) \end{array}$$

the vertical arrows are equivalences.

Proof. We have seen the existence and commutativity of the diagrams in Lemma 03EG. We have to check hypotheses (1) – (5) of Lemma 03A0 for the induced functor $u : \mathcal{C}/U \to \mathcal{D}/u(U)$. This is completely mechanical.

Property (1). This is Lemma 03EG.

Property (2). Let $\{U'_i/U \to U'/U\}_{i\in I}$ be a covering of U'/U in $\mathcal{C}/U$. Because u is continuous we see that $\{u(U'_i)/u(U) \to u(U')/u(U)\}_{i\in I}$ is a covering of $u(U')/u(U)$ in $\mathcal{D}/u(U)$. Hence (2) holds for $u : \mathcal{C}/U \to \mathcal{D}/u(U)$.

Property (3). Let $a, b : U''/U \to U'/U$ in $\mathcal{C}/U$ be morphisms such that $u(a) = u(b)$ in $\mathcal{D}/u(U)$. Because u satisfies (3) we see there exists a covering $\{f_i : U''_i \to U''\}$ in $\mathcal{C}$ such that $a \circ f_i = b \circ f_i$. This gives a covering $\{f_i : U''_i/U \to U''/U\}$ in $\mathcal{C}/U$ such that $a \circ f_i = b \circ f_i$. Hence (3) holds for $u : \mathcal{C}/U \to \mathcal{D}/u(U)$.

Property (4). Let $U''/U, U'/U \in \mathrm{Ob}(\mathcal{C}/U)$ and a morphism $c : u(U'')/u(U) \to u(U')/u(U)$ in $\mathcal{D}/u(U)$ be given. Because u satisfies property (4) there exists a covering $\{f_i : U''_i \to U''\}$ in $\mathcal{C}$ and morphisms $c_i : U''_i \to U'$ such that $u(c_i) = c \circ u(f_i)$. We think of U''_i as an object over U via the composition $U''_i \to U'' \to U$. It may not be true that c_i is a morphism over U! But since $u(c_i)$ is a morphism over $u(U)$ we may apply property (3) for u and find coverings $\{f_{ik} : U''_{ik} \to U''_i\}$ such that $c_{ik} = c_i \circ f_{ik} : U''_{ik} \to U'$ are morphisms over U. Hence $\{f_i \circ f_{ik} : U''_{ik}/U \to U''/U\}$ is a covering in $\mathcal{C}/U$ such that $u(c_{ik}) = c \circ u(f_{ik})$. Hence (4) holds for $u : \mathcal{C}/U \to \mathcal{D}/u(U)$.

Property (5). Let $h : V \to u(U)$ be an object of $\mathcal{D}/u(U)$. Because u satisfies property (5) there exists a covering $\{c_i : u(U_i) \to V\}$ in $\mathcal{D}$. By property (4) we can find coverings $\{f_{ij} : U_{ij} \to U_i\}$ and morphisms $c_{ij} : U_{ij} \to U$ such that $u(c_{ij}) = h \circ c_i \circ u(f_{ij})$. Hence $\{u(U_{ij})/u(U) \to V/u(U)\}$ is a covering in $\mathcal{D}/u(U)$ of the desired shape and we conclude that (5) holds for $u : \mathcal{C}/U \to \mathcal{D}/u(U)$. □